

Nicholas Lam

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EDUCATION

Honors in Computer Science (HBsc.), Honors in Business Administration (HBA) London, ON, Canada
Western University, Ivey Business School September 2020 - April 2025

- Western Scholarship of Distinction: \$2500, Dean's Honor List
- **Leadership:** Vice President of Projects Portfolio Western Founders Network Club, WesternAI Project Director
- **Awards:** DeltaHacksX Hackathon Winner, MapleHacks Hackathon Winner, Business 2257 Robert G. Siskind Entrepreneurial Award (1st/250), Simplii Financial Case Competition 1st Place, Relay Financial Start-up Advisory Program 1st Place

EXPERIENCE

Gather AI Pittsburgh, PA
Software Engineer II, Full-Stack March 2026 - Present
Software Engineer I, Full-Stack October 2024 - February 2026

- **Exception Review Platform:** Led development of a data labeling platform as part of an outsourcing initiative to audit **1,000+ drone-flagged exceptions daily**, freeing **60 hours/week** of internal QA capacity
- Designed **Cochran-formula statistical sampling** with deterministic seeding and per-tenant fairness guarantees, minimizing review volume while maintaining **90% confidence coverage** across all warehouse customers
- Built **real-time collaborative exception tagging** using Server-Sent Events and RxJS, enabling simultaneous multi-reviewer workflows with live state synchronization
- **Multi-Tenant Dashboard:** Designed and built a multi-tenant analytics dashboard that eliminated all manual metric compilation across facilities for the customer success team, serving inventory accuracy with period-over-period trend analysis
- Engineered a **nightly metrics precomputation pipeline** replacing on-the-fly SQL aggregations with cron-scheduled jobs, caching **12 metrics across 60 facilities** — drone flight performance, ML barcode accuracy, and volumetric pallet estimation — serving queries across all facilities for up to 3 months of data in **sub-second response times**
- Built **hierarchical multi-tenant authentication middleware** with Auth0 JWTs, enforcing user-subset-of-organization tenant scoping with fail-secure defaults and Azure Application Insights audit logging

Royal Bank of Canada - Amplify Program Toronto, ON, Canada
Software Engineer Intern May 2024 - August 2024

- Revamped RBC's Insurance Application Process to leverage **machine learning** predictive modeling reducing **client drop off rates by 10%** and increasing **straight-through processing rate by 40%** improving client retention and experience
- Engineered and deployed a robust, multilayered **Microservice API** with an orchestration layer to optimize performance, ensuring isolation for high availability and fault tolerance using **FastAPI and Docker** containerization and deployment
- Developed and implemented comprehensive **unit tests** for reliability, robustness, and scalability of each microservice
- Project was chosen to go forward for a **provisional patent** and is currently being implemented into the bank

PROJECTS

Nodes  | *Next.js, React, Node.js, Chrome Extension API, Gmail SMTP*

- Built **Nodes**, a platform automating professional networking, both sourcing and sending emails, with **1,000+ users**
- Developed **Chrome Extension** with recursive iframe scraping across 7 concurrent iframes, parallelizing LinkedIn page loads to process **100+ profiles in under 30 seconds**
- Implemented a **prefix tree** with incremental traversal for O(1) company-to-email format matching independent of dataset size across **500+ companies**, with longest-match resolution and early termination on invalid prefixes
- Engineered rate-limited email pipeline using Gmail SMTP with connection pooling, 20-person batch processing, and retry logic with exponential backoff

coTA - DeltaHacksX Winner  | *React, FastAPI, Python, WebSocket, Python-PPTX, Cohere RAG Model*

- Educational Media Platform that converts lecture material documents into short-form TikTok-like content
- Created a dual RESTful API and WebSocket backend with FastAPI to extract text from PowerPoint files, and feed them to Cohere's Retrieval Augmentation Generation (RAG) model that transforms them into summaries displayed on the front-end
- Dynamically fed data to the front-end through WebSockets reducing content generation time from 2 minutes to 15 seconds

TECHNICAL SKILLS

Languages: C++, C, Python, Java, TypeScript, JavaScript, SQL, HTML & CSS

Frameworks & Libraries: React, Next.js, Node.js, NestJS, Express.js, FastAPI, Flask, RxJS, Socket.io, TensorFlow

Databases & Infrastructure: PostgreSQL, MongoDB, Firebase, Docker, AWS, Azure, Auth0